

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

A New Metric for Tracking Frontier AI Autonomous Capability Growth

~7 months

50% Time Horizon
Doubling Time

$R^2 = 0.97$

6-Year Fit
2019–2025

GPT-2 → o3

2 sec → 2 hours
6 years of growth

Thomas Kwa, Ben West, Joel Becker, et al. — Model Evaluation & Threat Research (METR)

arxiv.org/abs/2503.14499 | March 2025

A New Metric: 50% Task-Completion Time Horizon

The 50% time horizon is the task duration (in human professional time) that an AI model can complete at a 50% success rate.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

STEP 1

Assemble Task Suite

170 diverse software/research tasks spanning 3 sec to 8 hrs of human effort

STEP 2

Collect Baselines

Time skilled humans & measure AI agent success rates on each task

STEP 3

Fit Logistic Model

Estimate task duration at which AI succeeds 50% of the time

Key advantage: human-grounded — compares AI directly to human task performance

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

The benchmark combines three curated task suites with over 800 human baselines totaling 2,529 hours of measurement.

HCAST

97 tasks | 1 min – 30 hrs

RE-Bench

7 tasks | All ~8 hrs

SWAA

66 tasks | 1 sec – 30 sec

Example Tasks by Duration

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: which file is a shell script?
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix bug in molecular dynamics simulation inputs
munge_data	56 minutes	Write Python script to transform JSON by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30× speedup of backtesting tool

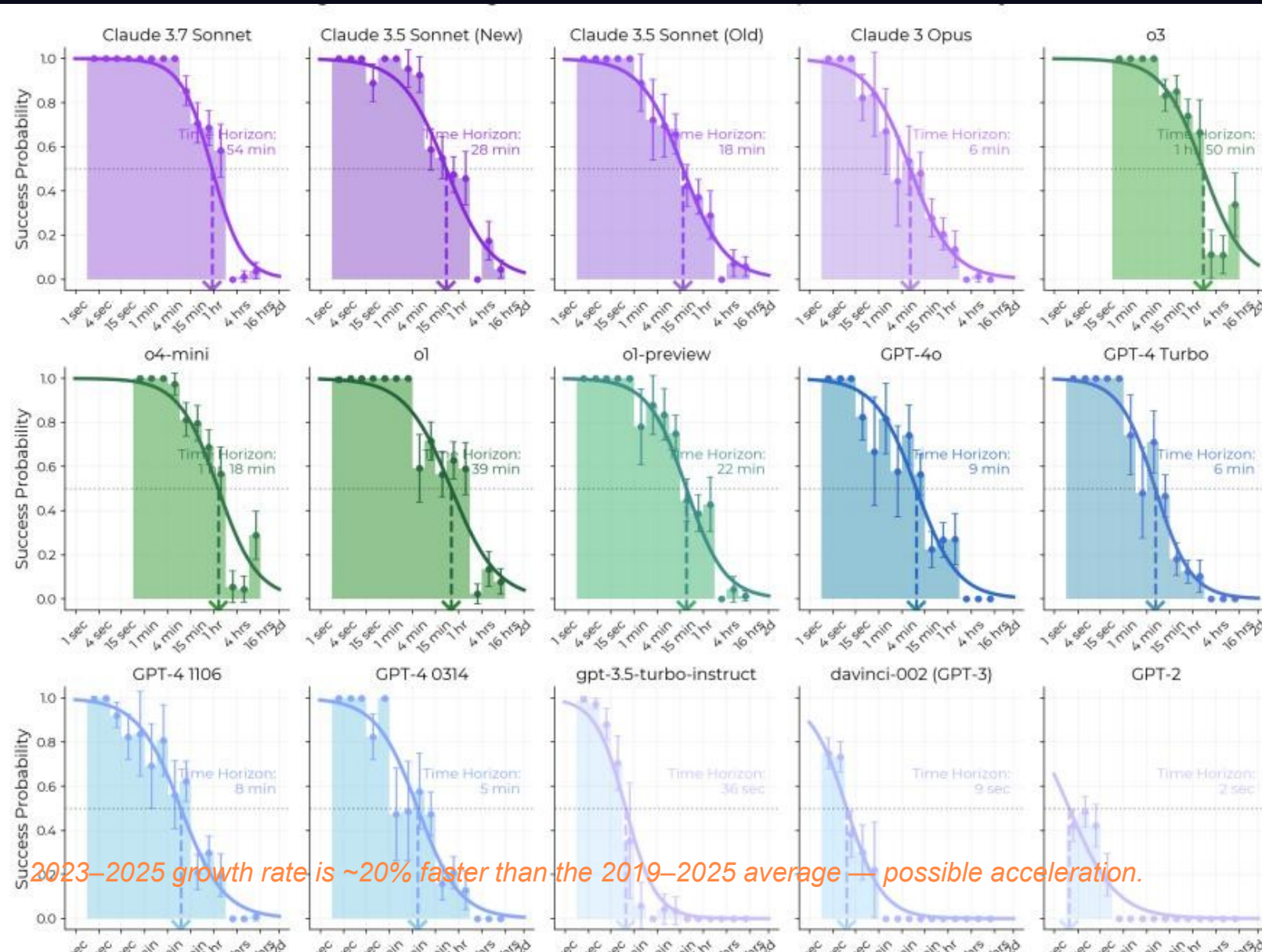
Measurement Scale

170 Total Tasks

800+ Human Baselines

2,529 hrs Human Time Measured

AI Task Horizon: 2 Seconds (GPT-2) to ~2 Hours (o3) in 6 Years



2023–2025 growth rate is ~20% faster than the 2019–2025 average — possible acceleration.

Doubling Time

~207 days
(~7 months)

Goodness of Fit

$R^2 = 0.97$

Key Milestones

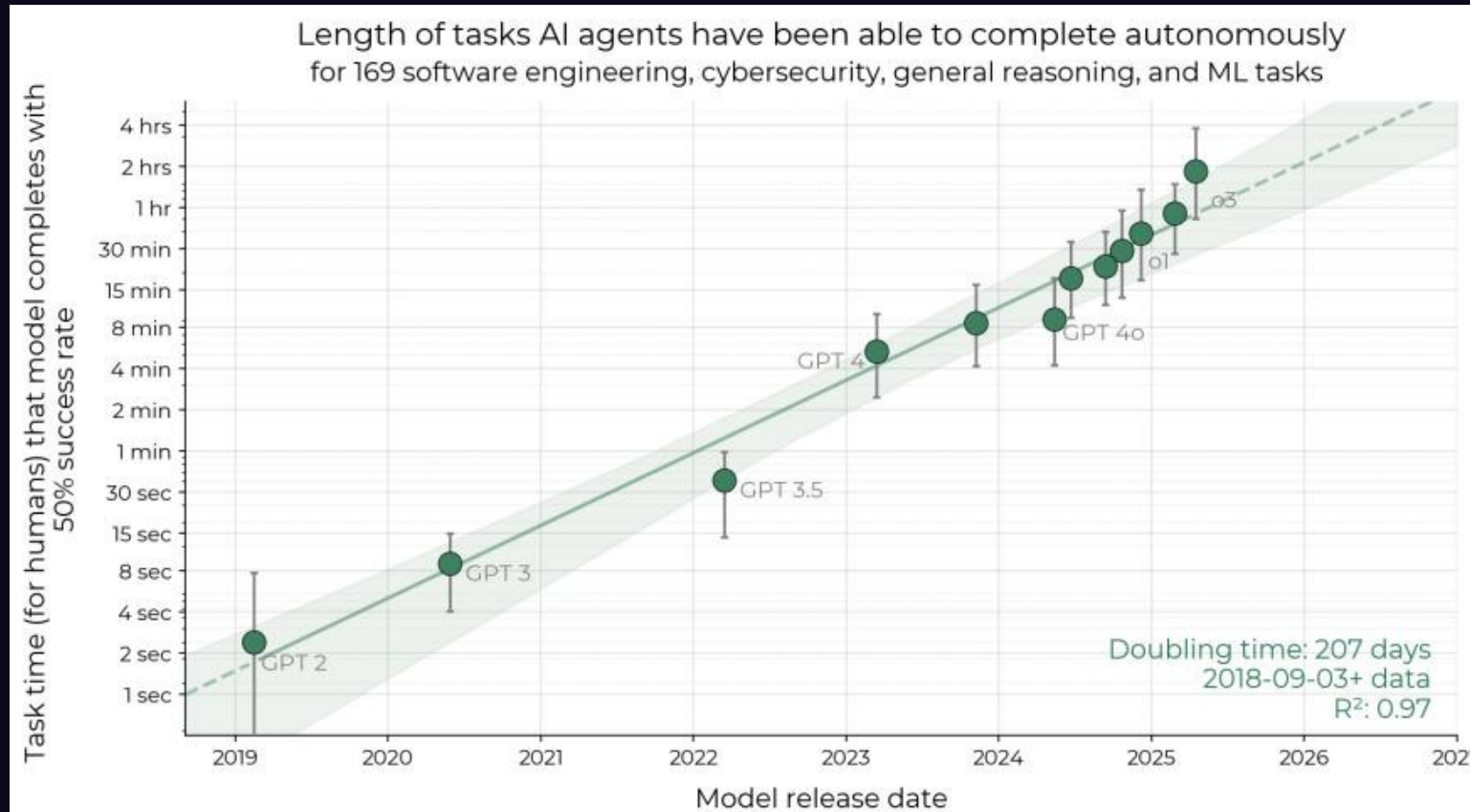
GPT-2 (2019) ~2 sec

GPT-4 (0314) (2023) ~5 min

o3 (2025) ~2 hrs

Success Probability Drops Sharply as Task Length Increases

Per-model success probability curves show how each model's success rate declines with task duration. Frontier models (Claude 3.7 Sonnet, o3) push the curve rightward significantly.



Interpretation

Left of curve

Short tasks (minutes) — frontier models succeed ~80–100%

Center of curve

Medium tasks (minutes–1 hr) — most differentiation occurs

Right of curve

Long tasks (hours) — even frontier models drop to ~50%

Claude 3.7 Sonnet reaches ~54 min horizon; o3 reaches ~1 hr 50 min — both far beyond GPT-2's ~2 sec.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Improved Logical Reasoning

Models can plan multi-step solutions and recover from dead ends without human intervention

Better Tool Use

Reliable execution of shell commands, file operations, code execution, and web search across extended tasks

Greater Reliability & Self-Awareness

Knowing when to stop, when to ask for clarification, and when to self-correct before wasting computation

Ability to Adapt to Mistakes

Detecting when a step has failed and dynamically replanning rather than blindly continuing

Limitation: Performance is notably lower on less structured, "messier" tasks. External validity remains an open question.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

If the trend continues, AI reaches a >1-month time horizon (167 work hours) between mid-2028 and mid-2031

Implying automation of many month-long software projects within the next 3–5 years

Growth Rate Comparison

2019–2025

Baseline rate

~207 days
doubling

2023–2025

~20% faster

Shorter doubling
time ~165 days

Caveats & Open Questions

- External validity: tasks may not perfectly represent average real-world intellectual labor
- Trend changes: economic/strategic decisions or capability plateau could alter trajectory
- 80% time horizon is ~5× shorter than 50% — fully reliable automation lags significantly
- Supplementary experiments found little evidence trends are slower on more realistic tasks

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1 50% Time Horizon is a Robust Metric

$R^2 = 0.97$ over 6 years of model releases — an intuitive, human-grounded measure of AI autonomy

2 Doubling Every ~7 Months with Possible Acceleration

2023–2025 growth is ~20% faster than the 2019–2025 average. o3 already at ~2 hrs of human-equivalent work

3 Frontier AI Now Completes ~2-Hour Tasks Autonomously

Up from 2 seconds for GPT-2 in 2019. Exponential trend shows no sign of plateauing in frontier models

4 External Validity Remains an Open Question

Tasks may not perfectly represent average real-world intellectual labor; supplementary experiments cautiously optimistic

5 Key Limitations: Unstructured Tasks & Generalization

Poor performance on "messier" tasks; uncertain how trends generalize to real-world job distributions